

Hardware-Efficient Three-Input Fuzzy MPPT for PV Battery Charger ICs

Mehmet Gök, Kemal Ozanoglu, and Günhan Dündar
Department of Electrical and Electronics Engineering, Bogazici University, Istanbul, Türkiye

Abstract—Maximizing power extraction from photovoltaic (PV) arrays under dynamic environmental conditions is critical for battery charging. Conventional MPPT (Maximum Power Point Tracking) algorithms suffer from slow transients and steady-state oscillations. To address these limits, this paper presents an ASIC-oriented FLC (fuzzy logic controller) modeled in MATLAB/Simulink using three parameters: ΔV_{pv} , ΔI_{out} , and V_{pv} . To avoid complex hardware division in the FLC, a hardware-efficient method utilizing look-up tables and bitwise shift operations is proposed. Simulation results show a steady-state PV tracking efficiency exceeding 99%, a buck converter efficiency of $\sim 95\%$, and millisecond-level transient recovery during sudden irradiance drops, achieving near-optimal performance with significantly reduced complexity.

Index Terms—Fuzzy Logic Control (FLC), Maximum Power Point Tracking (MPPT), Photovoltaic (PV) Systems, Multi-Input Fuzzy Inference, Battery Charging, MPPT Optimization

I. INTRODUCTION

The integration of photovoltaic (PV) systems is critical for sustainable power generation, but PV power output is non-linear and depends heavily on dynamic environmental conditions. MPPT algorithms are essential to maximize energy extraction, reduce energy waste, and improve battery charging efficiency.

Although conventional algorithms such as Perturb and Observe (P&O) and Incremental Conductance (IC) are widely used [8], they are often enhanced with intelligent control techniques such as fuzzy logic to handle rapidly changing conditions [6].

In this work, we introduce a hardware-efficient, three-input fuzzy MPPT architecture that combines a deterministic boundary safety override with multi-input fuzzy logic tracking. By processing ΔV_{pv} , ΔI_{out} , and V_{pv} using only simple look-up tables and bitwise shifts, the design eliminates complex arithmetic blocks, making it ideal for low-cost power management ICs. The system overview is shown in Figure 1.

The remainder of this paper is organized as follows: Section II details the system architecture including the PV model and buck converter, Section III describes the proposed fuzzy logic controller, Section IV presents the simulation results, and Section V concludes.

II. SYSTEM ARCHITECTURE

This project aims to charge a $\sim 12V$ car battery using PV modules that have a Maximum Power Point (MPP) voltage of $\sim 30V$. Because the PV voltage is significantly higher than the target battery voltage, a buck converter is chosen to efficiently step down the voltage. MATLAB/Simulink is chosen as the

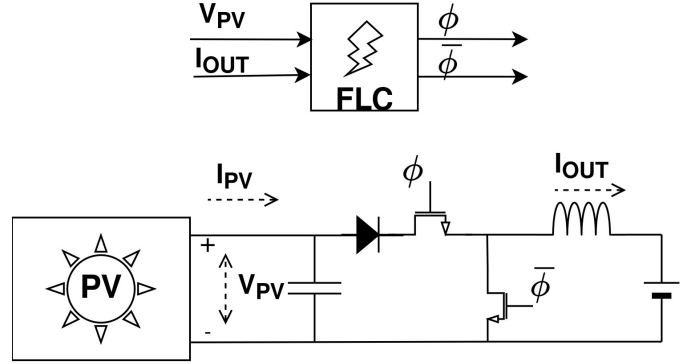


Fig. 1. Simulink model overview of the proposed fuzzy logic-based MPPT system for PV battery charger.

simulation environment for implementing and validating the proposed system (Figure 1).

A. Photovoltaic Panel Model

The PV model's example I-V & Power characteristics, based on De Soto et al. [1], are shown in Figure 2. As observed in the figure, despite varying irradiance conditions, the MPP is compressed into a narrow voltage range (around 30V).

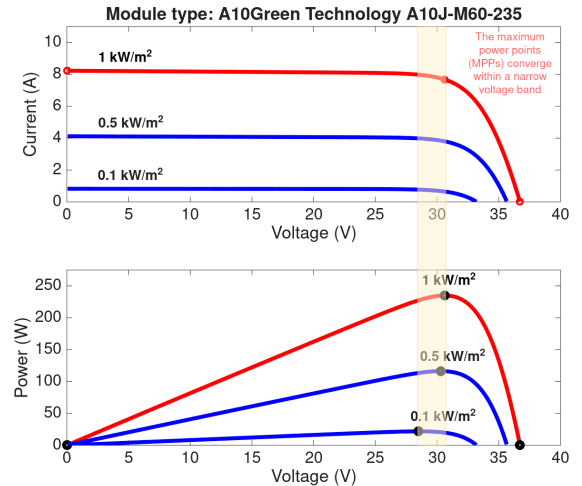


Fig. 2. I-V and power characteristics of the photovoltaic panel under various irradiance conditions.

Our tracking goal is to maintain system operation within this narrow target range. The controller adjusts the converter's

duty cycle to modulate the PV panel impedance and maintain optimal operation. The inference mechanism details are provided in Section III.

B. Buck Converter

1) *Buck Converter Specifications:* The buck converter specifications, derived from [2]–[4] and optimized for efficient power conversion, are presented in Table I. Parasitics of passive components are excluded in this system-level model.

TABLE I
BUCK CONVERTER SPECIFICATIONS

Parameter	Symbol	Value
Input Voltage (MPP)	V_{in}	30.6 V
Output Voltage (Battery)	V_{out}	14.4 V
Switching Frequency	f_{sw}	500 kHz
Sampling Frequency	f_{sample}	20 kHz
Inductance	L	2.6 μ H
Input Capacitance	C_{in}	47 μ F
Parasitic Resistance of Switch	R_{sw}	10 m Ω
Diode Forward Voltage	V_f	0.5 V

III. FUZZY LOGIC CONTROLLER

The fuzzy logic controller takes three input parameters (ΔV_{pv} , ΔI_{out} , and V_{pv}) to compute the duty cycle change (ΔD). The block diagram is shown in Figure 3.

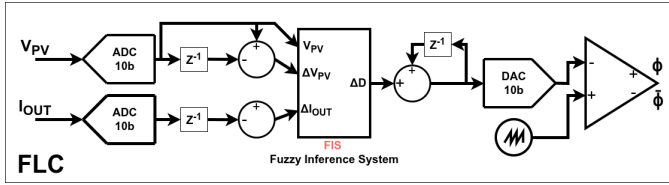


Fig. 3. Fuzzy logic controller block diagram showing the three phases of operation.

Input sampling and final duty adjustment are straightforward processes. Therefore, this section focuses on the key operations: inference, fuzzification, and defuzzification.

A. Inference

This phase evaluates IF-THEN rules using the **min** operator for logical AND. The inputs ΔV_{pv} and ΔI_{out} are mapped to three fuzzy sets: **Negative**, **Zero**, and **Positive**. V_{pv} is mapped to **Low_Voltage**, **Optimal_Voltage**, and **High_Voltage**. The output ΔD is mapped to five sets: **Big_Decrease**, **Decrease**, **Hold**, **Increase**, and **Big_Increase**.

3 inputs with 3 fuzzy sets would yield a total of 27 ($3 \times 3 \times 3$) rules. However, to reduce silicon area, we streamline this to 11 rules. There is another advantage to this approach: it prevents local minima trapping by prioritizing the absolute voltage level (V_{pv}) over the incremental changes (ΔV_{pv} and ΔI_{out}). When V_{pv} is outside the MPP range, two boundary safety rules override and suppress the others (Table II). When V_{pv} is optimal, only the 9 rules based on voltage and current variations are evaluated (Table III). This approach maintains the advantage of avoiding local minima while optimizing the

hardware implementation. Effectively FLC is constructed with P&O enhancement.

TABLE II
FUZZY LOGIC SAFETY RULES

PV Voltage (V_{pv})	Output (ΔD)
Low_Voltage	Big_Decrease
High_Voltage	Big_Increase

TABLE III
FUZZY LOGIC RULES FOR OPTIMAL RANGE

dI \ dV	Negative	Zero	Positive
Negative	Decrease	Hold	Increase
Zero	Hold	Hold	Hold
Positive	Increase	Hold	Decrease

B. Fuzzification

The hardware implementation utilizes three primary types of membership functions: Z-shaped, S-shaped, and Triangular-shaped. The degree of membership is found by mapping the crisp digital 10-bit input to a linear segment. The membership range value is chosen as 0 to 255 ($M = 255$). It is extracted using linear interpolation. To avoid complex division, we round the denominator to the nearest power of 2, which allows us to replace the division with a simple right bitwise shift. The membership calculation procedure is summarized in Algorithm 1.

Algorithm 1 Hardware-Efficient Linear Interpolation for Membership Functions

Require: Crisp input x , Thresholds A and B , Max Output M
Ensure: Membership degree $\mu \in [0, M]$

- 1: **if** $x \leq A$ **then**
- 2: $\mu \leftarrow 0$ {Or M for falling edge (Z-shape)}
- 3: **else if** $x \geq B$ **then**
- 4: $\mu \leftarrow M$ {Or 0 for falling edge (Z-shape)}
- 5: **else**
- 6: $\Delta x \leftarrow |x - A|$
- 7: $\Delta_{th} \leftarrow B - A$
- 8: $shift_val \leftarrow \text{LUT_Nearest_Power_Of_2}(\Delta_{th})$
- 9: $\mu \leftarrow (\Delta x \times M) \gg shift_val$
- 10: **end if**

To translate this algorithm into efficient Register-Transfer Level (RTL) logic, the hardware replaces the exact denominator ($B - A$) evaluation with a combinational look-up table. This table categorizes the interval width and maps it to its closest power of 2. The dynamic shift amount scales up to 11 ($2^{11} = 2048$) for wide fuzzy sets that span the full 10-bit ADC limit. However, to prevent excessive quantization errors and potential overflow when evaluating extremely narrow fuzzy boundaries, the power of 2 approximation deliberately stops near $2^4 = 16$ (a minimum shift amount of 4). Differences smaller than this lower bound automatically default to the

minimum shift constraint. During processing, the input distance (Δx) is multiplied by the maximum membership degree ($M = 255$)—a constant operation easily synthesized using bitwise shifts and a subtractor—and then logically right-shifted by the evaluated amount. This architecture entirely eliminates the need for complex, area-intensive hardware dividers.

All input parameters share a similar fuzzy set structure, utilizing Z-shaped, Triangular-shaped, and S-shaped functions with varying ranges to provide comprehensive coverage. An example of these RTL-emulated membership functions is shown in Figure 4.

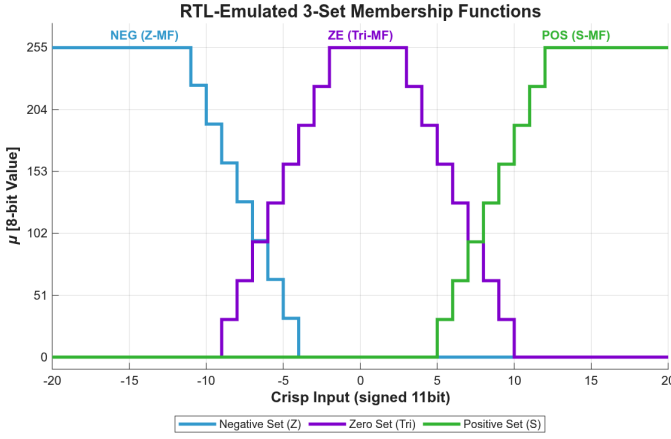


Fig. 4. RTL-emulated 3-set membership functions, illustrating the Z-shaped, Triangular-shaped, and S-shaped structures.

C. Defuzzification

A Sugeno-type weighted average defuzzifies the inference output into a crisp value, ΔD :

$$\Delta D = \frac{\sum_{i=1}^n w_i \cdot c_i}{\sum_{i=1}^n w_i} \quad (1)$$

Here, n is the active rule count, w_i is the firing strength, and c_i is the constant output coefficient. These c_i values are hardware-encoded constants defined in tabular form corresponding to the five output fuzzy sets: **Big_Decrease** (-16), **Decrease** (-4), **Hold** (0), **Increase** (4), and **Big_Increase** (16).

To eliminate the complex hardware division, a structure similar to Algorithm 1 is used. The denominator sum ($\sum w_i$) is mapped to its nearest power of 2, allowing the division to be replaced by an efficient bitwise right-shift. Furthermore, since the non-zero c_i values are strictly powers of 2, the multiplication ($w_i \cdot c_i$) is precisely executed via simple bitwise shift operations, completely eliminating the need for hardware multipliers.

IV. SIMULATION AND RESULTS

The performance of the proposed fuzzy logic-based MPPT controller was validated through a comprehensive simulation conducted in MATLAB/Simulink. The total simulation time was 0.6 seconds, divided into three distinct sections to demonstrate the controller's performance during both steady-state and transient conditions.

The PV efficiency, calculated as P_{pv}/P_{pvmax} , is the primary metric for evaluating the MPPT algorithm's effectiveness. The converter efficiency is evaluated as P_{bat}/P_{pv} . Due to the absence of signal filtering in the measurement blocks, instantaneous efficiency readings can momentarily appear above 100% during sharp transients.

Phase 1: Initial Stabilization (0s - 0.2s): During the first 0.2 seconds, the system operates with a manual, fixed duty cycle setting. The fuzzy logic controller is not active, allowing the converter to stabilize. The overall simulation results across the full 0.6 seconds, illustrating this initial phase, are depicted in Figure 5.

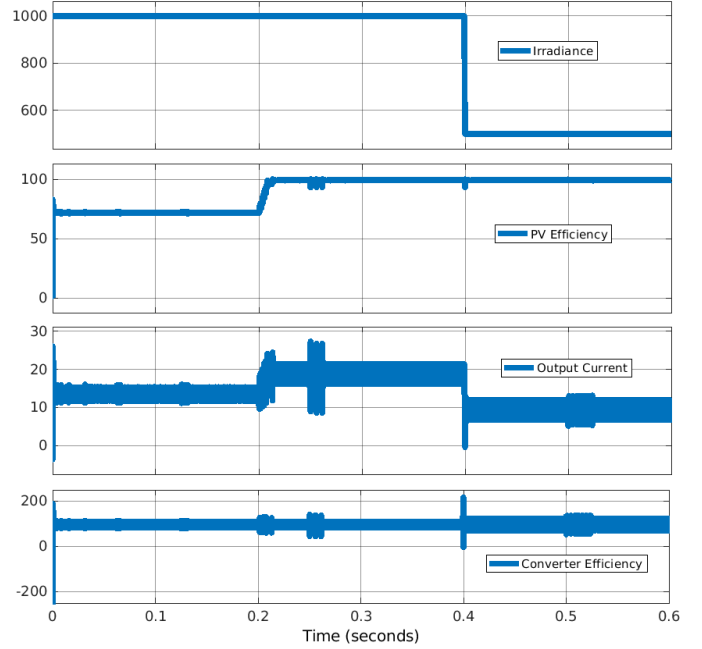


Fig. 5. Overall MATLAB simulation results across the three performance phases.

Phase 2: Fuzzy Controller Activation (0.2s): At 0.2 seconds, the fuzzy logic MPPT controller is activated. Figure 6 provides a zoomed-in view of this specific moment in time. The controller rapidly adjusts the duty cycle to bring the system to the maximum power point (MPP), achieving over 99% PV efficiency in just 13.4 milliseconds.

Phase 3: Transient Response (0.4s): At 0.4 seconds, the solar irradiance is dropped from 1000 W/m^2 to 500 W/m^2 to evaluate the effect of sudden environmental changes. Figure 7 shows the zoomed-in transient response. Despite the disturbance causing a momentary dip in PV efficiency, the fuzzy controller responds swiftly, restoring the efficiency to over 99% in approximately 1 millisecond.

These individual observations confirm the robustness and high performance of the fuzzy logic controller, showcasing its ability to quickly track the MPP and maintain high efficiency, even under rapidly changing environmental conditions.

Table IV summarizes the steady-state performance of the proposed system under the evaluated varying irradiance con-

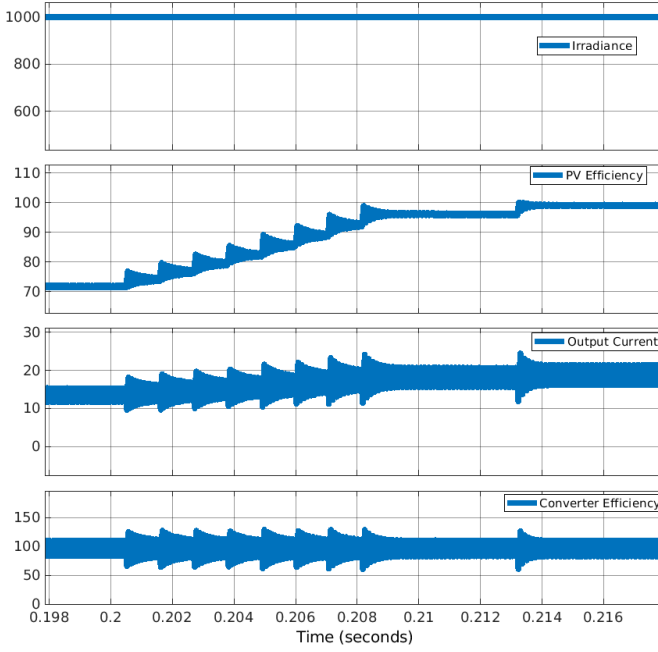


Fig. 6. Zoomed-in view of the FLC activation at 0.2s, showing rapid convergence to MPP.

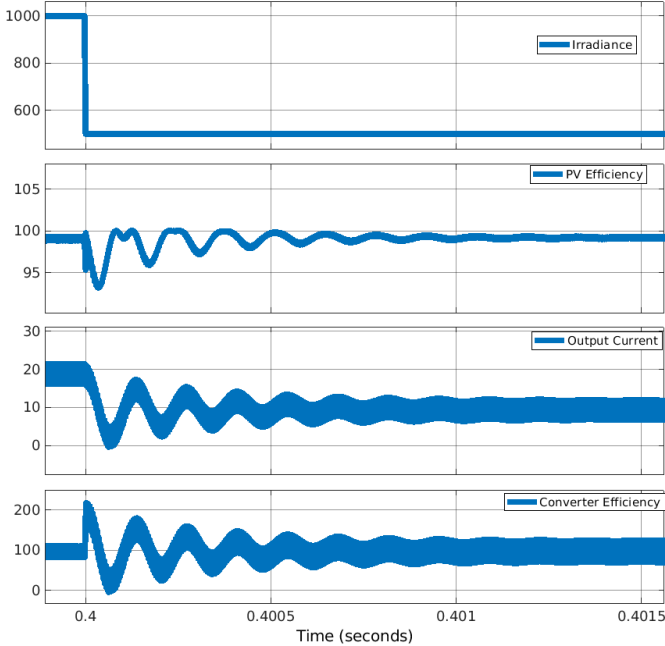


Fig. 7. Zoomed-in view of the transient response at 0.4s during an irradiance drop from 1000 to 500 W/m^2 .

ditions. The results demonstrate that the fuzzy logic controller maintains exceptional PV tracking efficiency while the buck converter sustains high power conversion efficiency across varying environmental states. Notably, the PV efficiency and buck converter efficiency remain around 99% and 95%, respectively, up until the converter enters Discontinuous Conduction Mode (DCM) at very low irradiance levels, which explains the efficiency drop observed at 200 W/m^2 .

TABLE IV
STEADY-STATE PERFORMANCE METRICS

Irradiance (W/m^2)	1000	500	200
PV Efficiency (%)	99.3	99.1	99.0
Buck Converter Efficiency (%)	95.2	94.9	78.8

V. CONCLUSIONS

This paper presents an ASIC-oriented, three-input fuzzy MPPT controller for PV battery chargers. By evaluating ΔV_{pv} , ΔI_{out} , and V_{pv} , the system avoids local minima and achieves highly responsive tracking. A key contribution is the hardware-efficient architecture, replacing complex multipliers and dividers with look-up tables and bitwise shifts to minimize IC area. Simulations confirm > 99% PV tracking efficiency and rapid transient recovery at 500 kHz switching frequency, outperforming traditional 2-input systems (Table V).

TABLE V
COMPARISON BETWEEN PREVIOUS 2-INPUT FLC AND PROPOSED 3-INPUT FLC

Parameter	Previous Work	Proposed Work
FLC Type	Mamdani	Sugeno
PV Tracking Efficiency	$\approx 97\%$	> 99%
Buck Conv. Efficiency	$\approx 96\%$	$\approx 95\%$
Switching Frequency	10 kHz	500 kHz
Settling Time	> 50ms	< 15ms
IC Area Optimization	Multiplier-based	LUT & Bitwise Shifts

This approach offers an optimal trade-off between accuracy and hardware complexity for low-cost ASICs. Future work targets physical IC implementation. Models and code are available at <https://github.com/mehmetgok1/MATLAB>.

REFERENCES

- [1] W. De Soto, S.A. Klein, and W.A. Beckman, "Improvement and validation of a model for photovoltaic array performance," *Solar Energy*, vol. 80, no. 1, pp. 78–88, 2006.
- [2] R. Erickson and D. Maksimovic, "Fundamentals of Power Electronics," 2nd ed. Springer, 2001.
- [3] M. H. Rashid, "Power Electronics Handbook," 4th ed. Butterworth-Heinemann, 2017.
- [4] Texas Instruments, "Basic Calculation of a Buck Converter," Application Report SLVA477B, 2015.
- [5] Z. Dalala, Z. U. Zahid, H. Y. Kanaan, N. M. L. Ha, and D. C. Habineza, "Design and real-time implementation of a novel photovoltaic MPPT system," *IEEE Trans. Sustain. Energy*, vol. 7, no. 3, pp. 1101–1115, Jul. 2016.
- [6] T. S. Ubale and S. V. Chaudhari, "Review on fuzzy logic based MPPT for photovoltaic systems," in *Proc. IEEE Int. Conf. Elect. Eng. Technol. (ICETECH)*, 2016, pp. 1–6.
- [7] L. A. Zadeh, "Fuzzy sets," *Inf. Control*, vol. 8, no. 3, pp. 338–353, 1965.
- [8] D. P. Hohm and M. E. Ropp, "Comparative study of maximum power point tracking algorithms," *Prog. Photovolt. Res. Appl.*, vol. 11, no. 1, pp. 47–62, 2003.